



The Role of Contextual Variation in Learning Cantonese Tones from Naturalistic Speech

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Abstract

A central question in speech acquisition is how infants learn contrastive sound categories from variable acoustic cues. The Distributional Learning Across Contexts (DLAC) proposal suggests that infants may identify contrastive sound dimensions by tracking distributions across contexts, with greater contextual variation signaling contrastive categories. Most research on acquisition of sound categories has focused on simple two-way vowel and consonant contrasts, leaving open how infants learn more complex or tonal contrasts. This study addresses these gaps by analyzing naturalistic speech to examine whether the ease of learning Cantonese's six tonal contrasts is linked to their variation across different contexts. By comparing our findings with existing acquisition data, we show that this proposal could predict which contrasts are easier or harder to learn, suggesting that in absence of invariant acoustic cues, infants may rely on contextual variation to acquire complex sound categories.

Index Terms: Cantonese, tone, sound category learning, distributional learning, acquisition, contextual variation

1. Introduction

Linguistically, two sounds are contrastive if they differentiate meaning—for example, English [l] and [r] distinguish words like “led” and “red”. However, even before infants have access to top-down information such as word meaning, they begin learning the sound categories of their language [1, 2]. This bottom-up learning process involves mapping continuous acoustic signals onto discrete sound categories, a process known as sound category learning. This raises a fundamental question: how do infants identify sound categories that are contrastive in their native language?

1.1. Distributional Learning (DL)

The *Distributional Learning* (DL) account proposes that infants act as “statisticians,” tracking speech sound distributions to infer contrastive categories [3]. If a probability density function along an acoustic dimension (e.g., duration) exhibits distinct peaks—indicating a bimodal or multimodal distribution—infants could identify this acoustic dimension as linguistically important, and interpret these peaks as separate sound categories. This mechanism has been supported by controlled experiments, where exposing infants to speech with bimodal distributions leads to formation of two categories [3, 4, 5, 6]. These studies demonstrate that infants are sensitive to distributional patterns when clear multimodality is present in the input. However, multimodality is rarely observed in naturalistic speech—the input infants receive in daily life—which is highly variable with significant acoustic overlap between sound cate-

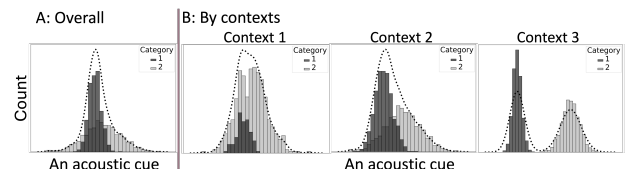


Figure 1: *Schematic demonstration on contextual variation.*

gories [7]. This lack of clear-cut multimodal distributions challenges the reliability of DL as a universal learning cue.

The *Distributional Learning Across Contexts* (DLAC) proposal [8], as an extension to the DL model, emphasizes contextual variation in addition to static distributional peaks. Figure 1 schematically demonstrates the concept of contextual variation: even when the overall frequency distribution is unimodal, different contexts (e.g. prosodic position, neighboring sounds, etc) reveal distinct distributional shapes. Hitczenko and Feldman [8] suggests that infants identify contrastive acoustic dimensions by tracking how distributional shapes vary across different contexts. By comparing vowel length in naturalistic speech corpora from Japanese (a language with a vowel length contrast) and French (a language without one), the authors found that contrastive dimensions exhibit greater contextual variation. In other words, when an acoustic dimension is contrastive, its distributional shape changes more drastically across contexts.

Most research on distributional learning has focused on simple two-way sound contrasts in vowels and consonants. For example, studies have examined English /t/ vs. /d/ [3], English /l/ vs. /r/ [9], Dutch /a/ vs. /a/ [10], and Japanese long vs. short vowels [8]. However, two key gaps remain. First, little is known about how infants acquire multiple contrasts that are cued by a single acoustic feature. Within this dimension, contrasts may also vary in learnability, with some contrasts being relatively easy or hard to acquire. There is limited understanding of what contributes to this difference in learnability. This gap limits our ability to test a key prediction of the DLAC framework: within the same acoustic dimension, contrasts with greater contextual variation should be easier to learn. Second, despite the prevalence of tone languages, especially in Asia and Africa, the role of DL in lexical tone acquisition remains largely unexplored. Cantonese, with its complex tone system and well-documented learning asymmetries, provides an ideal test case. The following section explores how these asymmetries make Cantonese tones particularly suitable for investigating the DLAC.

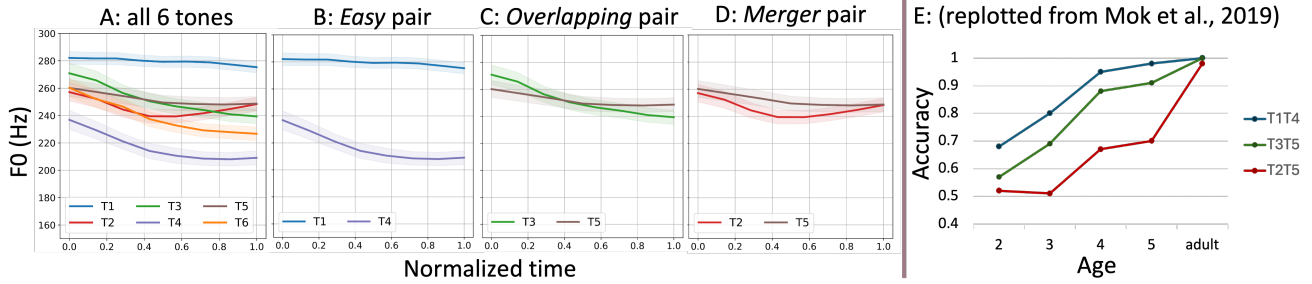


Figure 2: *F0* contours and developmental learning trajectories of Cantonese tone pairs. (A-D) Time-normalized pitch contours with mean and 95% confidence intervals from a female speaker (3519 tokens). (A) All six tones. (B) T1T4 shows clear phonetic separation, while (C) T3T5 and (D) T2T5 exhibit increasing degrees of acoustic overlap. Data from the MeHKCC corpus [11]. (E) Developmental learning trajectories for these tone pairs, replotted from [12].

1.2. Cantonese tones: a learning puzzle

Cantonese is a variety of Chinese primarily spoken in Guangdong, southern Guangxi, Hong Kong, Macau, and beyond. It is characterized by a complex lexical tone system with six distinct tones: high-level (T1), high-rising (T2), mid-level (T3), low-falling (T4), low-rising (T5), and low-level (T6). Each tone distinguishes word meaning; for example, *si* can mean “poetry” (*si1*), “history” (*si2*), “to try” (*si3*), “time” (*si4*), “market” (*si5*), or “be” (*si6*). Learning Cantonese tones can be challenging because children must distinguish between six tones that differ primarily in a single acoustic cue—pitch (measured with fundamental frequency, *F0*). In controlled experiments, tones could be differentiated by their pitch height and pitch contour [13]. However, naturalistic speech often blurs these distinctions. As illustrated in Figure 2, which shows a speaker’s tones in naturalistic speech, most tones are highly overlapping.

Despite these challenges, children eventually master Cantonese tones, though some tone pairs are harder to learn than others [12, 14, 15]. Fig. 2E illustrates children’s accuracy in distinguishing three different tone pairs at various ages [12]. The T1 vs. T4 contrast is learned more quickly and accurately than other pairs, whereas the T2 vs. T5 contrast is acquired the latest. The T2 vs. T5 distinction is particularly noteworthy as it is undergoing a sound change and merging into one category, similar to T3 vs. T6 and T4 vs. T6 [16, 17]. These developmental patterns, along with acoustic salience and sound change status, allow tone pairs to be categorized into three groups:

- *Easy pairs* – Pairs involving T1 (e.g., T1T4). These are acoustically distinct and learned early. T1 is the most acoustically salient, making it the easiest to distinguish from other tones, a pattern observed in both perception experiments and machine classification studies [13].
- *Merger pairs* – Pairs undergoing merger (e.g., T2T5, T3T6, T4T6). These exhibit extensive acoustic overlap and are not fully learned until adulthood.
- *Overlapping pairs* – All other pairs (e.g., T3T5). These exhibit acoustic overlap but remain learnable with experience.

Crucially, while both *Overlapping* and *Merger* pairs exhibit acoustic overlap, *Overlapping* pairs are more learnable. This raises a key question: Why are some acoustically overlapped tone pairs easier to learn than others?

This study takes an initial step toward addressing this puzzle by testing whether the DLAC framework could explain these learning asymmetries. Specifically, this study investigates whether the ease of acquiring different tonal contrasts is linked

to their distributional variation across contexts. By analyzing three selected tone pairs and comparing our findings with existing age-of-acquisition and perception data, we show that the DLAC framework could predict which contrasts should be easier or harder to learn. These findings suggest that, in the absence of invariant acoustic cues, infants may rely on contextual variation to acquire complex sound categories.

Given that Cantonese tone discrimination involves 15 binary contrasts, we selected three representative tone pairs from each group for detailed analysis: T1T4 as an *Easy* pair, T3T5 as an *Overlapping* pair, and T2T5 as a *Merger* pair. The T1T4 pair was chosen because it consists of the highest and lowest tones in Cantonese, making it the most distinct and easily separable contrast. The T2T5 pair was selected because it represents the most studied and well-documented tone merger in Cantonese [16]. The T3T5 pair was chosen due to its acoustic overlap and its shared T5 tone with the T2T5 pair, making it an ideal case for comparison with the T2T5 merger.

2. Method

2.1. Data

We analyzed the Multi-ethnic Hong Kong Cantonese Corpus (MeHKCC) [11], a corpus of annotated child-directed speech (CDS) and adult-directed speech (ADS) produced by mothers in Hong Kong with diverse linguistic backgrounds. The data were originally collected between 2019 and 2021 and include recordings from various speech tasks, such as film retelling, one-on-one interviews, map descriptions, story retelling, word identification, and in-home recordings. For this study, we selected only native Cantonese-speaking mothers, resulting in a dataset of naturalistic speech from 25 female speakers. We extracted monosyllabic and disyllabic words with unambiguous tonal annotations while excluding function words. Although the corpus includes both CDS and ADS, we do not distinguish between them in the current analysis; however, future work may explore potential differences between these speech registers. We extracted the following information about each syllable produced by the mothers.

2.1.1. Acoustic cues

We extracted nine *F0* descriptors per syllable to capture both pitch height and the temporal dynamics of the *F0* contour: mean *F0*, median *F0*, variance of the *F0* contour, *F0* range (maximum–minimum), onset *F0*, 25% *F0*, 75% *F0*, offset *F0*, and *F0* contour duration. To minimize pitch perturbations at syllable

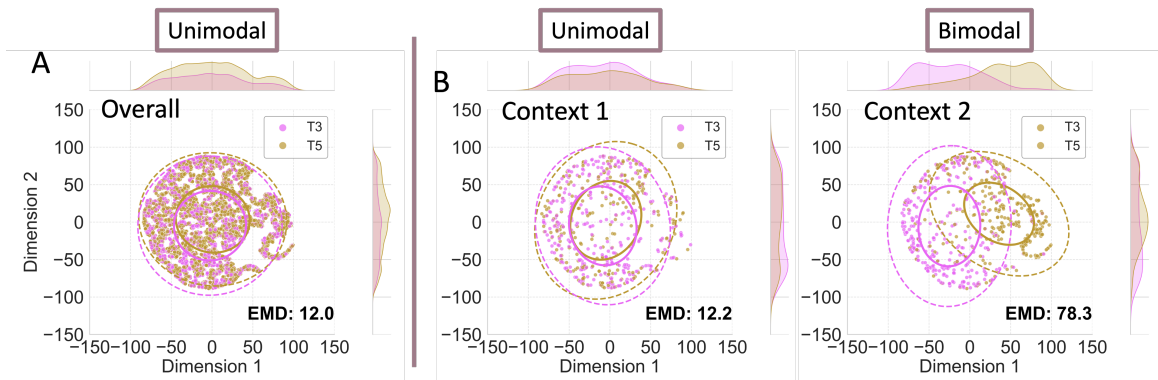


Figure 3: (A) The overall frequency distribution of the Overlapping tone pair T3T5 along a reduced 2D space shows a unimodal distribution, despite T3 and T5 being contrastive tones. (B) Frequency distributions for the same tone pair across two specific contexts (e.g. Context 2 = nasal onsets without codas, second syllable, utterance-medial) reveal different distributional shapes.

edges, we excluded the first and last 15% of each syllable. Following previous work [17], pitch values for each speaker were normalized relative to their mean F0 for T1, a stable level tone. Outliers were removed by filtering syllables exceeding 2.5 standard deviations from the mean for each F0 descriptor, resulting in 59,385 syllables for analysis (T1: 17,147; T2: 9,339; T3: 7,277; T4: 8,622; T5: 4,167; T6: 12,833).

2.1.2. Contexts

Each syllable was categorized based on four contextual factors known to influence the realization of Cantonese tones [18, 17, 19]: (1) the class of onset sounds, which includes 8 categories (e.g., unaspirated stops, fricatives, nasals); (2) the class of coda sounds, which is divided into 3 categories (empty coda, nasals, stops); (3) the syllable's position within a word (monosyllabic words, first syllable of disyllabic words, second syllable of disyllabic words); and (4) its prosodic position (utterance-initial, utterance-medial, or utterance-final). These factors were also used in [8].

2.2. Distributional analysis

Unlike previous studies that rely on a single acoustic measure (e.g. duration, VOT), our approach captures the dynamic use of F0 with nine descriptors. Given the high dimensionality, we employed t-distributed Stochastic Neighbor Embedding (t-SNE) via the *sklearn* package in Python to reduce the data to a two-dimensional space for visualization and analysis [20]. t-SNE is an unsupervised nonlinear dimensionality reduction technique that embeds high-dimensional data in a low-dimensional space in such a way that similar objects are modeled by nearby points and dissimilar objects are modeled by distant points in the space. In the visualization of the t-SNE results, we overlaid ellipses representing one and two standard deviations around the mean of each tonal distribution to highlight the spread and variation of the data. Additionally, we included marginal density plots along the x- and y-axes to illustrate the distributional shapes in each dimension.

Following [8], we also measured the differences between tonal distributions using Earth Mover's Distance (EMD) [21], also known as the Wasserstein distance, using the *scipy* package in Python. EMD quantifies the effort required to transform one distribution into another by computing the optimal transport cost between them. This metric is particularly well-suited for

comparing probability distributions that may differ in both location and shape. We computed EMD separately for the marginal densities along the x and y axes of the t-SNE results, summing them to obtain the final EMD measure: $EMD_{total} = EMD_x + EMD_y$. Higher EMD values indicate greater distributional separation. This process is repeated for each specific context. We report the mean, median, maximum, and variance of EMD across contexts for each tone pair. Higher mean EMD values indicate generally greater distributional separation between two tones, while higher variance reflects greater contextual variation.

We predict that the separability and degree of contextual variation in distributional shapes will align with developmental acquisition patterns. Specifically, *Easy* contrasts (e.g., T1T4) will exhibit the highest overall separation and the greatest variation across contexts, as reflected in the highest EMD mean and variance. *Overlapping* contrasts (e.g., T3T5) will show moderate separation and variation. *Merger* contrasts (e.g., T2T5) will have the least distributional separation and the lowest variation, reflecting in the lowest EMD mean and variance.

3. Results

Figure 3 visualizes the frequency distribution of the *Overlapping* tone pair, T3T5, after dimensionality reduction. Panel A reveals an overall unimodal distribution for the pair, despite the fact that T3 and T5 are contrastive tones. This suggests that, when aggregated across contexts, the tonal distinction may not always be easily perceptible. However, Panel B presents a more nuanced view by illustrating the distributional shapes of the T3T5 pair in two specific contexts. In Context 2—characterized by nasal onsets without codas, second-syllable position within a word, and utterance-medial position—the tonal distributions exhibit greater separability, as reflected in a higher Earth Mover's Distance ($EMD = 78.3$). In contrast, in Context 1—defined by the absence of onsets and codas, with the same syllable and prosodic position—the distributions show much lower separability ($EMD = 12.2$). This comparison suggests that certain contexts, such as nasal onsets, enhance the distinctiveness of tone contrasts.

To further investigate whether contextual variation aids in sound category learning, we examined whether the degree of contextual variation in the distributional shapes aligns with developmental acquisition patterns. Figure 4 presents a boxplot

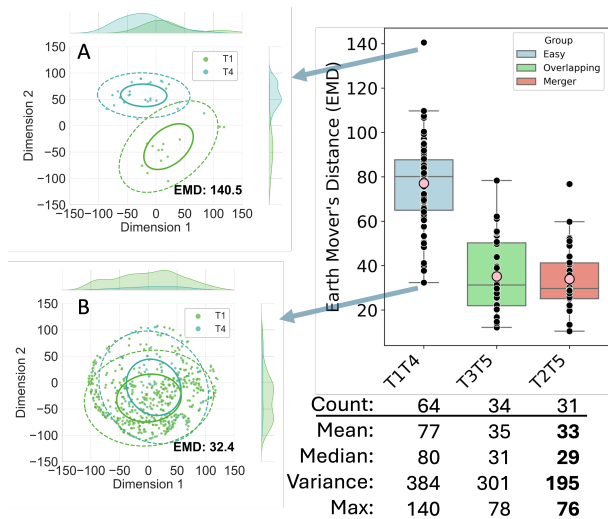


Figure 4: *EMD distributions for three tone pair categories: Easy (T1T4), Overlapping (T3T5), and Merger (T2T5). Panel A illustrates a context with the greatest separability for T1T4, while Panel B shows a context with the lowest separability. Lowest values for four EMD metrics (mean, median, variance, and maximum) were bolded.*

of EMD values for the three tone pairs across all contexts. Each data point represents the pairwise EMD value between two tones within a single context. Panels A and B highlight two specific distributions of the T1T4 pair, one with a particularly high EMD value and one with the lowest EMD value, emphasizing the range of distributional variation within different contexts. Greater overall distributional separation is reflected in higher mean, median, and maximum EMD values and greater contextual variation is reflected in higher variance. The specific metric values are shown in Figure 4.

The *Easy* pair (T1T4), which has more data points than the other two pairs, exhibits high mean, median, and maximum EMD values, indicating that T1 and T4 remain well-separated overall regardless of contexts. This pattern aligns with their acoustic distinctiveness observed in Figure 2. In contrast, the *Overlapping* pair (T3T5) and the *Merger* pair (T2T5) exhibit comparably low mean, median, and maximum EMD values, suggesting that both pairs are generally less separable than T1T4. The key distinction between the *Overlapping* pair (T3T5) and the *Merger* pair (T2T5) lies in the EMD variance: T3T5 shows substantially higher variance than T2T5, indicating that the distributional distance between T3 and T5 changes more drastically depending on the context, whereas the distance between T2 and T5 remains consistently similar and low across all contexts.

These results support the hypothesis that greater contextual distributional variation facilitates tone learning, as predicted by the DLAC framework. The *Easy* tone pair (T1T4) exhibits the most distinct separation, suggesting it should be easier for infants to learn. The *Overlapping* pair (T3T5) shows less separability but relatively high contextual variation, implying that while this contrast may be more challenging, it remains learnable with experience. Finally, the *Merger* pair (T2T5) exhibits the least separation and least contextual variation, suggesting the greatest difficulty. These patterns align with the developmental data presented in Figure 2E.

4. Discussion

This study investigated the DLAC with the specific question: whether the ease of learning sound contrasts is influenced by the degree of contextual variation in the input. Using three Cantonese tone pairs in naturalistic speech as a test case, our findings revealed that tone pairs that are acquired more readily exhibit greater distributional separation and contextual variation than those that are more challenging to learn. These current results provide empirical support for the generalizability of the DLAC framework to both complex sound contrasts and tonal systems, underscoring the framework’s potential as a mechanism for learning in the absence of invariance in speech signals.

An underlying assumption of this work is that infants learn tones in a pairwise manner. Our focus on pairwise comparisons was driven by two reasons. First, we want to maintain consistency with prior DL research which primarily examines two-way contrasts. Second, our study aligns its predictions with developmental data, which rely on pairwise discrimination paradigms. However, our choice does not imply that infants necessarily learn tones exclusively through pairwise contrasts. It remains possible that learners acquire all six tones simultaneously or that they first establish a reference tone (e.g., T1) before differentiating others. Future research could employ computational modeling to simulate different learning mechanisms and compare their predictions to empirical data.

This study extends prior research by examining a different stage of sound category learning than was studied in prior work. [8] applied the DLAC framework to explain the initial step—identifying the relevant acoustic dimension that differentiates contrastive sounds (e.g. duration, F0). We focus on the next step: how learners identify contrastive categories along an already-identified relevant acoustic dimension. In other words, we assume that infants have already recognized F0 as the key acoustic cue but have yet to learn the tone categories. By investigating this later stage, our study strengthens the generalizability of the DLAC proposal, demonstrating that it serves as a comprehensive learning account that explains both stages of sound category acquisition. Future research could focus on the earlier stage of acquisition, and study how Cantonese-learning children identify F0 as a relevant acoustic cue. One way to do this would be to compare the distribution of F0 against dimensions that are non-contrastive in Cantonese, such as duration. Alternatively, a cross-linguistic comparison between Cantonese and a non-tonal language like English—where F0 does not encode lexical contrasts—could also provide further insights.

In summary, our results suggest that rather than hindering sound category learning, contextual variation in distributional shapes may serve as a cue that helps infants distinguish meaningful contrasts, even in the absence of clearly separable multimodal distributions assumed in traditional DL models. A natural next step for this research is to analyze the full set of tone pairs and compare their distributional properties with developmental data. There is also room for methodological exploration. For instance, alternative dimensionality reduction techniques such as principal component analysis, or different distributional distance metrics like Kullback–Leibler (KL) divergence, could be employed. Future work could also explore different F0 descriptors as well as additional contexts (e.g. including lexical information). Additionally, future experimental work could also provide direct evidence that complements the current study. For example, participants could be exposed to varying levels of contextual distributional variation, and assessed on how this variation impacts their ability to learn sound categories.

5. References

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